

ALL DSA PATTERNS with LEETCODE PROBLEMS

1. Two Pointers **Pattern**

👉 Used when dealing with arrays / strings, usually sorted

Key idea: move two indices based on condition

Problems:

- LC 1 – Two Sum
 - LC 125 – Valid Palindrome
 - LC 167 – Two Sum II
 - LC 344 – Reverse String
 - LC 11 – Container With Most Water
 - LC 15 – 3Sum
 - LC 42 – Trapping Rain Water
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2. Sliding Window **Pattern**

👉 Fixed / variable size subarray or substring problems

Key idea: expand & shrink window

Problems:

- LC 3 – Longest Substring Without Repeating Characters

- LC 76 – Minimum Window Substring
 - LC 209 – Minimum Size Subarray Sum
 - LC 567 – Permutation in String
 - LC 438 – Find All Anagrams
 - LC 1004 – Max Consecutive Ones III
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3. Prefix Sum **Pattern**

👉 Fast range queries

Key idea: precompute cumulative sums

Problems:

- LC 303 – Range Sum Query
 - LC 304 – Range Sum Query 2D
 - LC 560 – Subarray Sum Equals K
 - LC 974 – Subarray Sums Divisible by K
 - LC 525 – Contiguous Array
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4. Hashing (Map / Set) **Pattern**

👉 Frequency counting, fast lookup

Problems:

- LC 1 – Two Sum
- LC 49 – Group Anagrams

- LC 128 – Longest Consecutive Sequence
 - LC 242 – Valid Anagram
 - LC 347 – Top K Frequent Elements
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5. Fast & Slow Pointers **Pattern**

👉 Cycle detection

Problems:

- LC 141 – Linked List Cycle
 - LC 142 – Linked List Cycle II
 - LC 876 – Middle of Linked List
 - LC 202 – Happy Number
 - LC 234 – Palindrome Linked List
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6. Binary Search **Pattern**

👉 Sorted arrays / monotonic answers

Problems:

- LC 704 – Binary Search
- LC 33 – Search in Rotated Sorted Array
- LC 34 – First & Last Position
- LC 153 – Find Minimum in Rotated Sorted Array
- LC 875 – Koko Eating Bananas

- LC 1011 – Capacity To Ship Packages
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7. Monotonic Stack **Pattern**

👉 Next Greater / Smaller Element problems

Problems:

- LC 496 – Next Greater Element I
 - LC 503 – Next Greater Element II
 - LC 739 – Daily Temperatures
 - LC 84 – Largest Rectangle in Histogram
 - LC 85 – Maximal Rectangle
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8. Heap / Priority Queue **Pattern**

👉 Top K, scheduling

Problems:

- LC 215 – Kth Largest Element
 - LC 347 – Top K Frequent Elements
 - LC 295 – Find Median from Data Stream
 - LC 621 – Task Scheduler
 - LC 973 – K Closest Points
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9. Greedy **Pattern**

👉 Local optimal → global optimal

Problems:

- LC 55 – Jump Game
 - LC 45 – Jump Game II
 - LC 121 – Best Time to Buy and Sell Stock
 - LC 134 – Gas Station
 - LC 435 – Non-overlapping Intervals
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10. Intervals **Pattern**

👉 Merge / overlap problems

Problems:

- LC 56 – Merge Intervals
 - LC 57 – Insert Interval
 - LC 252 – Meeting Rooms
 - LC 253 – Meeting Rooms II
 - LC 986 – Interval List Intersections
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11. Linked List **Pattern**

👉 Pointer manipulation

Problems:

- LC 206 – Reverse Linked List
 - LC 92 – Reverse Linked List II
 - LC 19 – Remove Nth Node
 - LC 21 – Merge Two Sorted Lists
 - LC 143 – Reorder List
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12. Tree DFS **Pattern**

👉 Depth-first traversal

Problems:

- LC 104 – Max Depth of Binary Tree
 - LC 543 – Diameter of Binary Tree
 - LC 110 – Balanced Binary Tree
 - LC 124 – Binary Tree Maximum Path Sum
 - LC 236 – Lowest Common Ancestor
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13. Tree BFS **Pattern**

👉 Level-order traversal

Problems:

- LC 102 – Level Order Traversal
- LC 103 – Zigzag Level Order
- LC 199 – Right Side View

- LC 515 – Largest Value in Each Row
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14. Backtracking **Pattern**

👉 Explore all possibilities

Problems:

- LC 46 – Permutations
 - LC 47 – Permutations II
 - LC 39 – Combination Sum
 - LC 78 – Subsets
 - LC 131 – Palindrome Partitioning
 - LC 51 – N-Queens
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15. Dynamic Programming (1D) **Pattern**

👉 Optimal substructure

Problems:

- LC 70 – Climbing Stairs
 - LC 198 – House Robber
 - LC 53 – Maximum Subarray (Kadane)
 - LC 322 – Coin Change
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16. Dynamic Programming (2D) **Pattern**

👉 Grid / string DP

Problems:

- LC 62 – Unique Paths
 - LC 64 – Minimum Path Sum
 - LC 72 – Edit Distance
 - LC 1143 – Longest Common Subsequence
 - LC 300 – Longest Increasing Subsequence
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17. Graph (DFS / BFS) **Pattern**

👉 Connectivity problems

Problems:

- LC 200 – Number of Islands
 - LC 695 – Max Area of Island
 - LC 133 – Clone Graph
 - LC 994 – Rotting Oranges
 - LC 127 – Word Ladder
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18. Union Find (DSU) **Pattern**

👉 Connected components

Problems:

- LC 547 – Number of Provinces
 - LC 684 – Redundant Connection
 - LC 721 – Accounts Merge
 - LC 323 – Number of Connected Components
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19. Bit Manipulation **Pattern**

👉 XOR tricks

Problems:

- LC 136 – Single Number
 - LC 191 – Number of 1 Bits
 - LC 338 – Counting Bits
 - LC 268 – Missing Number
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20. Maths **Pattern**

👉 Mathematical logic

Problems:

- LC 50 – Pow(x, n)
- LC 172 – Factorial Trailing Zeroes
- LC 9 – Palindrome Number